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Five-year survival analysis of occlusal and proximal restorations performed in non-compliant children

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ABSTRACT

Objectives: To evaluate the survival of occlusal and proximal restorations in deciduous molars of non-compliant children treated under midazolam or nitrous oxide/oxygen (N_2O/O_2) and evaluate predictors of failure.
Methods: A total of 151 children (517 teeth) received occlusal (102 patients; 228 teeth) or proximal (105 patients; 289 teeth) restorations—and were assessed over 5 years. Kaplan–Meier survival and multilevel Cox proportional hazards models were applied, assessing patient-level (level 2) and tooth-level (level 1) variables—age, sex, restorative need, sedation type, cooperation, jaw, tooth side, isolation, and material.
Results: 16 children (60 teeth) received N_2O/O_2, 112 children (373 teeth) midazolam, and 23 children (84 teeth) no sedation. Sedation method was not related to restoration failures. Estimated survival at 48 months was 50 % for occlusal, and 60 % at 60 months for proximal restorations ($p = 0.072$). Occlusal failures were more likely in younger children ($p < 0.05$), lower jaw ($p = 0.046$), and with glass ionomer cements (GIC; $p = 0.028$), while proximal restorations failures occurred more likely in lower molars ($p = 0.024$) and with GIC ($p = 0.003$). Overall, predictors for occlusal and proximal failures were: younger age (HR = 0.27, $p = 0.030$); lower jaw (HR = 2.09, $p = 0.001$); and GIC (HR = 3.72, $p < 0.001$).
Conclusions: Neither restoration type nor sedation method were significant factors. Overall, restorations in non-compliant children are more likely to fail when placed in younger patients, in lower molars, or when glass ionomer cements are used.
Clinical relevance: Although sedation can help non-compliant children, these patients can still be difficult to treat. This study highlights that neither cavity type nor sedation method predicts restoration failures; instead, patient age, tooth location, and restorative material are key factors. These findings can help clinicians make evidence-based choices to improve treatment outcomes in pediatric patients requiring sedation.

1. Introduction

Dental anxiety and uncooperative behaviour are common challenges in paediatric dentistry, with over 10 % of children presenting behaviour management problems [1]. To help manage these situations, the European Association of Paediatric Dentistry (EAPD) proposes different sedation techniques like oral midazolam and nitrous oxide/oxygen (N_2O/O_2) inhalation [2]. Midazolam is a short-acting benzodiazepine, offers a combination of anxiolytic, sedative, and amnesic effects, and in doses between 0.25 mg/kg to 1 mg/kg in children it is associated with more cooperative behaviours [3], though it has a slow recovery. N_2O/O_2, on the other hand, still has low quality evidence, but it may provide mild sedation and pain control, so the EAPD recommends N_2O/O_2 as a conscious sedation agent only because of its well-known overall safety, its anxiolytic and sedative effects combined with its rapid onset and recovery [2]. Both these sedation methods are generally considered safe and effective, especially for children who are unable to undergo dental care using only behavioural management techniques. As such, sedation has become an important tool for paediatric dentists when treating uncooperative children. Although sedation techniques should not have any direct impact on restoration survivals, the behaviour of the sedated children are not totally guaranteed. Any difficult

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behaviour can negatively impact the dentists performance, so, there is still limited information considering their impact on the outcome of dental restorations.

Occlusal and proximal restorations are some of the most frequently performed procedures in primary molars affected by caries. The survival of these restorations depends on a multitude of factors, including the type of restorative material, the operator's technique, and individual patient-related factors like oral hygiene and caries risk [4]. Occlusal restorations tend to last longer than proximal, probably because of its simpler cavity shapes and lower stress from occlusal forces [5]. Composite restorations usually show higher survival rates than other materials. One study assessing composite, resin-modified glass ionomer cement (RMGIC) and conventional glass ionomer restorations in primary molars up to 4 years has shown survival rates of 62.2 %, 51.2 % and 48.4 %, respectively [6]. However, they did not differentiate the type of restoration (occlusal or proximal). In contrary, no difference between RMGIC and composite up to 2 years of follow-up had been observed, with a failure rate of 16 % [7]. Atramautic restorative treatment (ART) is often performed in uncooperative children, but it shows lower success rate than conventional treatment [8]. Knowing how long occlusal and proximal restorations typically last in primary teeth is important not only for guiding treatment choices but also for setting realistic expectations when planning long-term dental care for young patients.

Although sedation may lighten the load of treating uncooperative children, it still comes with its own challenges—such as limited treatment time, difficulty maintaining isolation, and unpredictable patient cooperation — all this can affect the final quality of the restorations. Moreover, children with behavioural problems or dental fear also present greater risk of caries [9], which, in turn, may reduce restoration longevity. We hereby question how treating children under midazolam or N_2O/O_2 sedation could impact the survival of occlusal and proximal restorations on deciduous teeth, with a follow-up period of up to five years. We also evaluate predictors of failure for the restorations.

2. Methods

2.1. Study design

This study was a single-center, practice-based, non-interventional, retrospective clinical study. All restorations and follow-up assessments were made during routine clinical care and recall appointments. All information gathered for this study was extracted from patient files, after treatment and follow-ups were concluded. According to guidelines for good clinical practice (Clinical trials – Directive 2001/20/EC) and the European guidelines for good clinical practice (CPMP/ICH/135/95) this study was, therefore, not subject to Medical Ethical Committee approval [10].

Reporting follows STROBE guidelines for cohort studies [11].

2.2. Sample selection

For this study, we considered as eligibility criteria: children of either sex who had undergone restorative treatment at the university clinic, on at least one deciduous molar, and that at least one of the restorations had been performed under either nitrous oxide / oxygen (N_2O/O_2) or midazolam sedation. It was not necessary that all teeth had to be treated under sedation, but at least one.

From all patients (aged 0 to 15 years old) treated at the Pediatric Dentistry Clinic of the University of Bern between the years of 2011 and 2023, we identified a total of 373 children (aged up to 10 years old) who required restorative treatment under sedation.

From this initial pool, we excluded cases of treatments on incisors and canines, treatments on permanent teeth, and cases of trauma or pulpotomy. Only cases of caries requiring occlusal and proximal restorations were included. We obtained complete clinical data for 151 patients, encompassing 517 deciduous molars (60 treated under N_2O/O_2, 373 under midazolam, and 84 under no sedation): from which, there were 228 occlusal restorations (in 102 patients) and 289 proximal restorations (in 105 patients).

2.3. Restorations

The Pediatric Dentistry Clinic serves as a referral centre for paediatric patients who present behavioural issues and cannot undergo treatment at their local general dentists. All restorations were made by specialist paediatric dentists who either already have completed their specialization or who are in their final year of specialization in Paediatric Dentistry.

The patients were treated either with midazolam (0.5 mg/kg; maximum dose of 10 mg) or with N_2O/O_2 (up to 40 %/60 %). The selection of one sedation type or the other was based on age of the patient (usually younger patients were treated with midazolam and older and/or heavier patients were treated with N_2O/O_2), their acceptance of the mask and after discussing both techniques together with the parents (parental consent).

The restorations were made either with cotton rolls (relative) or under total isolation (rubber dam or Isovac™ / Isolite System), and with composite, bullk-fill or glass ionomer. At the end of the treatment, the dentists usually give a subjective assessment of the patient's cooperation/behaviour during the treatment, as either good, moderate or bad.

2.4. Data extraction

A list of patients was identified from the accounting department of our clinic, where the code for sedation was used. From the list, two undergraduate students (UAC and JS) searched the patient files and extracted the data related to the dental treatments. The pseudo-anonymzed data was extracted by giving each patient a code. Data on patient level and tooth level were obtained. Patient-related data included: age of the child at the time of treatment, sex, restorative need, sedation type, cooperation. Tooth-related data included: location of restoration (occlusal or proximal), side of the mouth, upper or lower jaw, type of isolation and restoration material.

2.5. Survival outcome

During routine recall appointments, the patients were observed and their dentition status noted. Time in months from restoration until the final recall appointment was registered and survival curves up to 60 months (5 years) were generated. For the Kaplan-Meier survival analyses (which are event-focused), the original treatments were either considered as events (failures) or censored. *Failures* were considered whenever the restorations needed re-treatment (another restoration, pulpotomy, extraction, etc.). In such cases, the treatment was performed in follow-up clinical sessions and the new treatment was not included in the study. Also, the reason for failures (whenever written in the patient file) was noted. *Censored* restorations were considered when the original treatment was not marked as failure: e.g., the restoration was still present and successful, or the tooth exfoliated before the restoration experienced a failure, or the follow-up period ended before the restoration experienced a failure, or the patient dropped out of the study and the restoration was not assessed.

2.6. Statistical analyses

Initially, descriptive analyses characterised the cohorts. Then, Kaplan-Meier survival analyses compared the survival of occlusal and proximal restorations, calculating the cumulative survival probabilities, annual failure rates (AFR) and mean time-to-failure (in months) of each type of restoration. The AFR was calculated using the exponential model: $(1-y)^z = (1-x)$, where y is the average AFR throughout the 5

years, x the total failure rate at the end of the study, and z is time in years.

Later, bivariate and multilevel regressions were made. For the bivariate analyses, we used Cox regressions, where associations between the time-to-failure (outcome) and each independent variable was verified. The independent variables were grouped according to two levels: Tooth and Patient. The variables in the Tooth level were divided into Characteristics of the Restoration and Characteristics of the Tooth, including: isolation (cotton roll/total isolation), restoration material (composite/bulk-fill/glass ionomer cement), side of mouth (right/left), and jaw (upper/lower). The variables in the Patient level were divided into Characteristics of the Child and Characteristics of the Treatment, including: sex (female/male), age (0–2/3–4/5–6/7–8/9+), restorative need (low = 1 or 2 teeth; high = 3 or more teeth in need of restorative treatment), sedation (none/nitrous oxide/midazolam), cooperation (good/moderate/bad).

For the multilevel regressions, we verified the interplay of many different independent variables on the outcome. This multilevel model (Fig. 1) accounts for the fact that some restorations/teeth (level 1) were nested within the same patient (level 2) [12], and the variables from level 2 can influence (directly or indirectly) the variables in level 1 as well as the outcome [13]. For this, Cox Proportional Hazards Models were used. To account for unobserved heterogeneity in the baseline hazard of the patients (due to clustering of observations – multiple teeth per patient), we included frailty terms (random effects) at both Tooth (1st) and Patient (2nd) levels assuming gamma distributions. Separate multivariate models were made for occlusal and proximal restorations. The covariates were included in the model from distal (2nd Level) to proximal (1st Level) using a backward stepwise hierarchical approach. Significant covariates ($p < 0.05$) were retained in the final regression model. Hazard ratios (HR) were calculated for each covariate to estimate the relative risk of restoration failure. A HR greater than 1 indicates an increased risk of failure, while a HR <1 suggests a protective effect.

Models were made for occlusal and proximal restorations separately, and then an overall model (considering all restorations) was made. The descriptive and bivariate Cox analyses were performed in IBM SPSS Statistics (version 22), while the Kaplan-Meier curves and multilevel hierarchical Cox Hazards Regressions were modelled in R (version 4.4.3).

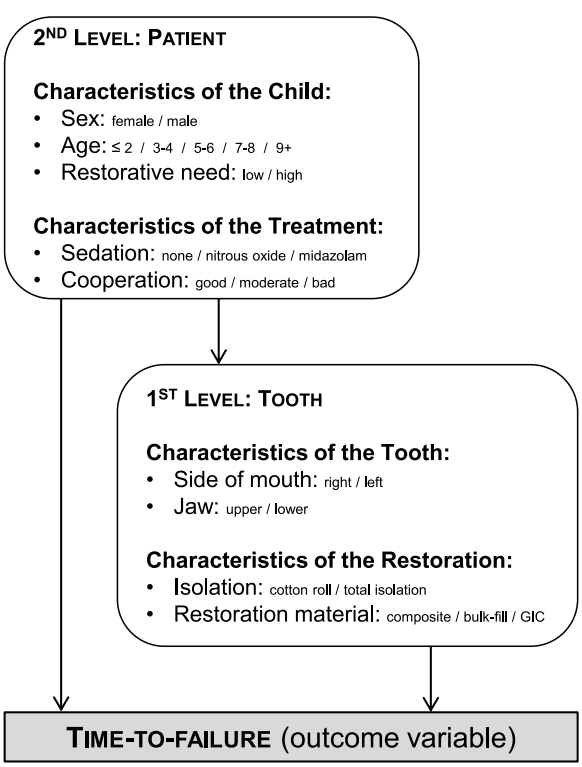

Fig. 1. Simplified conceptual hierarchical framework for the multilevel regression modeling.

3. Results

A total of 151 patients, encompassing 517 deciduous molars, were included, from which 16 children (60 teeth) were treated under N_2O/O_2, 112 children (373 teeth) under midazolam, and 23 children (84 teeth) without sedation. From the 517 teeth, 228 received a occlusal restoration, while 289 teeth received a proximal restoration.

Of the 228 occlusal restorations, a total of 53 failed over 5 years, with a mean time-to-failure of 41.3 ± 1.8 months and an AFR of 16.3 %. The estimated median survival time (50 % survival probability) for occlusal restorations was 48 months (4 years). For the 289 proximal restorations, 51 failed, with a mean time-to-failure of 46.5 ± 1.6 months and an AFR of 11.5 %. The estimated median survival time (50 % survival probability) for proximal restorations was >60 months (59.8 % at 5 years). There was no significant difference between occlusal and proximal restorations ($p = 0.072$; Fig. 2).

3.1. Occlusal restorations

Occlusal restorations (Table 1) were mostly performed in younger children (82 %) in age group 3–6 years old, under midazolam sedation (78 %), under relative isolation with cotton roll (69 %), and using either composite (26 %) or bulk-fill (56 %). Simple bivariate analyses showed that age may play a significant role on the survival of restorations, with each older age group having significantly lower risk of failure (45 % lower risk for 3–4 yo, 22 % for 5–6 yo, and 6 % for 7–8 yo). Also, children with bad behaviour during the treatment, as well as the restorations made with GIC, all had twice higher risk of failure.

Grouping the children into low and high need of restorative treatment, no significance difference ($p = 0.240$) was observed in the longevity of restorations (Fig. 3), despite low need patients show slightly more positive results.

In the multilevel analyses, accounting for the effect of the patients, the mixed-effects Cox model showed significant patient-level variability ($\sigma = 0.88$, variance = 0.78; $p = 0.006$). When the covariates were added to the model, variance decreased ($\sigma = 0.58$, variance = 0.34), indicating explanatory power from these covariates. The analyses (Table 1) showed that older children had significantly lower risk of failure: 5–6-year-olds (HR = 0.29, $p = 0.041$) and 7–8-year-olds (HR = 0.08, $p = 0.027$). A significantly greater risk of failure was found for teeth in the lower jaw (HR = 1.95, $p = 0.046$), and when GIC was used as restorative material (HR = 2.54, $p = 0.028$). This final model (including age, jaw and material) had an improved fit (integrated log-likelihood: $\chi^2 = 30.58$, $p < 0.001$; penalized log-likelihood: $\chi^2 = 55.75$, $p < 0.001$), confirming that these 3 predictors play a significant role on the hazard risks of occlusal restorations.

3.2. Proximal restorations

Proximal restorations were performed mostly in slightly older children (79 %) between 5–8 yo, under midazolam sedation (57 %) or nitrous oxide (22 %), under total isolation using rubber dam or suction isolation technology (62 %), and using either composite (38 %) or bulk-fill (55 %). Simple bivariate analyses showed that cooperation has a borderline association to failure ($p = 0.091$), but teeth from the lower jaw ($p = 0.024$) have a 91 % greater risk of failure, and those restored with GIC have three times higher risk of failure ($p = 0.003$).

Grouping the children according to the restorative need, no significance difference ($p = 0.450$) was observed (Fig. 4).

In the multilevel analyses, accounting for the effect of the patients, the mixed-effects Cox model revealed no significant patient-level

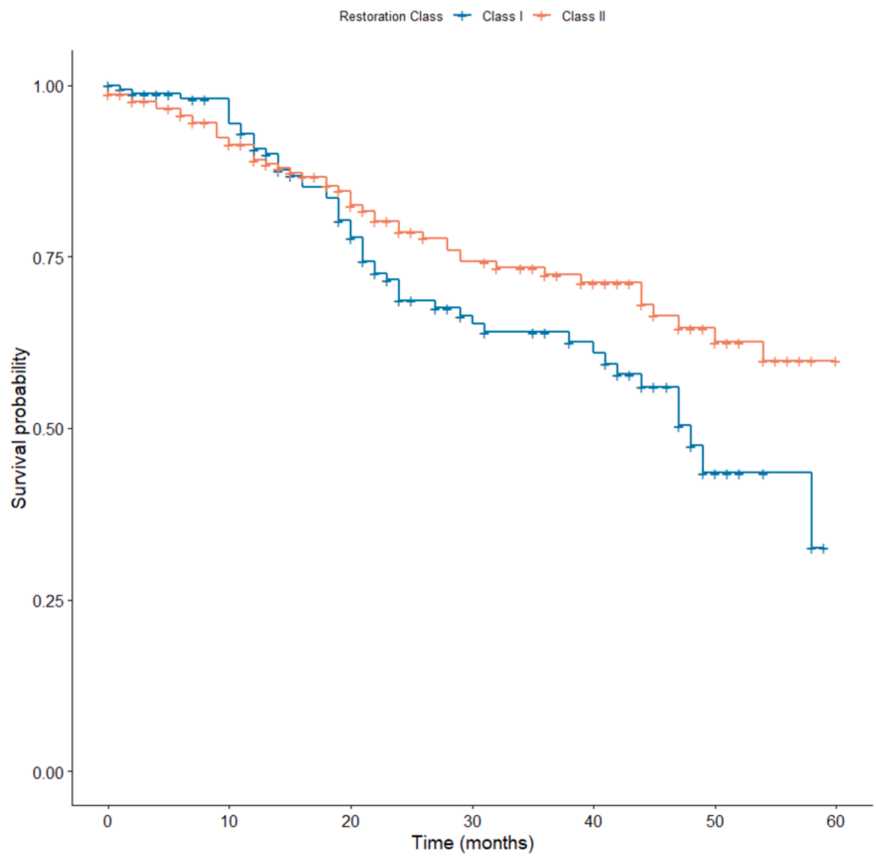

Fig. 2. Kaplan-Meier survival estimates for occlusal and proximal restorations during the 5 years of observation (p = 0.072).

variability in the baseline model ($\sigma = 0.68$, variance = 0.46, $p = 0.112$), but when the covariates related to the teeth were added to the model, there was a significant reduction in variance ($\sigma = 0.66$, variance = 0.44, $p < 0.001$). This indicates that the tooth-related covariates contribute significantly to the explanatory power of the model. The analyses showed (Table 2) that teeth from the lower jaw had twice the risk of failure compared to upper jaw teeth (HR = 2.51, $p = 0.003$) and GIC restorations had five times greater risk of failure (HR = 5.65, $p < 0.001$). This final model (containing only Jaw and Material) had an improved fit (integrated log-likelihood: $\chi^2 = 19.52$, $p < 0.001$; penalized log-likelihood: $\chi^2 = 51.14$, $p < 0.001$), confirming that these two covariates play a more significant role to the risk of failure of proximal restorations.

3.3. Reasons for failure

Looking at the reasons for failures (Table 3), we observed that a greater number of failures of occlusal restorations were due to either secondary caries or caries in a new surface of the tooth (77 %). Loss of the restorations was also a considerable reason (19 %), where 6 out of the 10 failures had been in restorations made with GIC. The reasons for failures of proximal restorations were also related to caries, though slightly lower (57 %) than occlusal. More failures were, however, related to pulpal involvement (18 %) and loss of restorations (22 %); the latter being mostly related to composite or bulk-fill restorations, and only 2 out of the 11 restorations lost due to fracture/loss were made with GIC.

3.4. Occlusal and proximal overall model

The multilevel analysis of occlusal and proximal restorations together (Table 4) showed moderate patient-level variability ($p < 0.001$) that improved with covariate adjustments. Restoration type did not significantly impact restoration failure (HR = 1.35, $p = 0.251$). The key predictors of failure included: age, where the older age group (7–8 years) had lower risk of failure (HR = 0.27, $p = 0.030$); jaw location, teeth in the lower jaw were at higher risk (HR = 2.09, $p = 0.001$); and restorative material, where GIC had increased risk of failure (HR = 3.72, $p < 0.001$). Cooperation level was borderline ($p = 0.093$), and like all other factors was not significant. This overall model had an improved fit ($\chi^2 = 62.71$, $p < 0.001$) over the baseline model, confirming the importance of these covariates in restoration failure.

4. Discussion

Our study is one of the first to specifically focus on the survival of restorations placed in non-compliant children, in whom the challenging behaviour makes performing high-quality restorative procedures considerably more difficult. Interestingly, our cohort showed a slightly higher survival probability for proximal restorations of 60 % at 5 years with AFR of 12 % than for occlusal restorations, where values were 50 % at 4 years and AFR of 16 %, but no significant difference between the two kinds of restorations ($p = 0.072$). Bearing in mind that the vast majority of the studies reporting survival rates of restorations in deciduous teeth are performed in compliant children, our values are remarkably within the range observed in the literature [5,14,15], where AFR values can reach up to 30 % [4].

To better understand our results, it is essential to consider the clinical context of our patients. The paediatric dentistry clinic at our department serves as a referral clinic for general dentists and private practices within a radius of over 20 km, where children with behavioural problems, who are not able to be treated at their local practices, are referred to our specialized team for treatment under sedation. Such patients usually have a sub-optimal oral hygiene, several caries lesions, and high risk of

Table 1
Bivariate and Multilevel analyses for occlusal restorations.

Independent Variables	Restorations		Bivariate analyses		Multivariate analysis	
	Censored n (%)	Failure n (%)	Hazard ratio (95 %CI)	p-value	Hazard ratio (95 %CI)	p-value
Level 2						
Characteristics of the Child						
Sex						
Females	81 (36)	26 (11)				
Males	94 (41)	27 (12)	0.91 (0.53–1.56)	0.719		
Age						
0–2	10 (4)	7 (3)				
3–4	78 (34)	35 (15)	0.45(0.20–1.04)	0.062	0.56(0.19–1.61)	0.282
5–6	65 (29)	10 (4)	0.22(0.83–0.59)	**0.003**	0.29(0.09–0.95)	**0.041**
7–8	20 (9)	1 (1)	0.06(0.01–0.52)	**0.010**	0.08(0.01–0.75)	**0.027**
9+	2 (1)	0 (0)	00(00–00)	0.983	00(00–00)	0.999
Restorative need						
Low	34 (89)	4 (11)				
High	141 (74)	49 (26)	1.83(0.56–5.95)	0.315		
Characteristics of the Treatment						
Sedation						
None	31 (14)	9 (4)				
N_2O/O_2	10 (4)	0 (0)	00(00–00)	0.971		
Midazolam	134 (59)	44 (19)	0.98(0.48–2.02)	0.961		
Cooperation						
Good	109 (48)	27 (12)				
Moderate	29 (13)	9 (4)	1.17(0.55–2.51)	0.683		
Bad	37 (16)	17 (7)	2.44(1.31–4.54)	**0.005**		
Level 1						
Characteristics of the Tooth						
Side of mouth						
Right	87 (38)	27 (12)				
Left	88 (39)	26 (11)	1.06(0.62–1.82)	0.832		
Jaw						
Upper	85 (37)	22 (10)				
Lower	90 (39)	31 (14)	1.38(0.79–2.38)	0.256	1.95(1.01–3.75)	**0.046**
Characteristics of the Restoration						
Isolation						
Cotton Roll	117 (51)	40 (18)				
Total isolation*	58 (25)	13 (6)	0.68(0.36–1.28)	0.233		
Restorative Material						
Composite	43 (19)	16 (7)				
Bulk fill	106 (46)	20 (10)	0.87(0.45–1.68)	0.672	0.74(0.35–1.58)	0.434
Glass Ionomer	26 (11)	17 (7)	2.74(1.38–5.45)	**0.004**	2.54(1.11–5.82)	**0.028**

* Rubber dam or suction isolation technology.

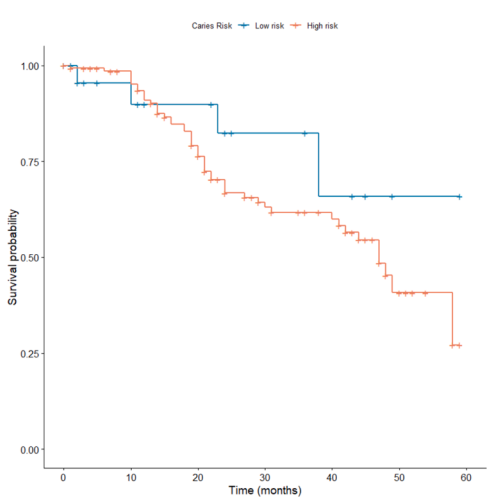

Fig. 3. Klapan-Meier survival estimates for occlusal restorations grouped by restorative need of the children, during the 5 years of observation (p = 0.240).

caries [16]. So, we speculate that the baseline success rate of any treatment for these children is already impacted by their background behaviour problems and sub-optimal oral hygiene [17].

This was somewhat reflected in our results, where despite grouping the baseline need of restoration into low (1 or 2 teeth) or high (≥3 teeth) restorative need [18], this variable did not impact survival rate. This finding is not surprising, as uncooperative children are also more likely to have broader behavioural and emotional difficulties at home or in everyday settings [19,20]. Such children typically fail to adhere to consistent home-care oral hygiene routines, leaving them persistently at high risk for restoration failures despite ongoing follow-up [21]. Furthermore, once these high-risk patients are sedated, they still remain awake and somewhat responsive (conscious sedation), leaving dentists with limited treatment time and unpredictable cooperation despite sedation. As a result, priority must be given to the most urgent treatment needs, while the evaluation of hygiene and any related indices is relegated to a secondary consideration — even at subsequent recall visits, patients frequently continue to be uncooperative, and thorough hygiene assessments may still be troublesome. So, in our patients, oral hygiene was only "assessed" subjectively, where the dentists simply state whether the hygiene is good or bad according to the presence of visible (non-stained) plaque. Due to this high variability and subjectivity, we were not able to include any hygiene data in our analyses.

Although we observed no differences between the survival of occlusal and proximal restorations in our cohort, we still analysed both restorations separately. For occlusal, the bivariate analyses showed a strong influence of patient-related and tooth-related factors. These restorations were mostly performed in a younger age group, from 3- to 6-year-olds. Despite the sedation with midazolam in these younger

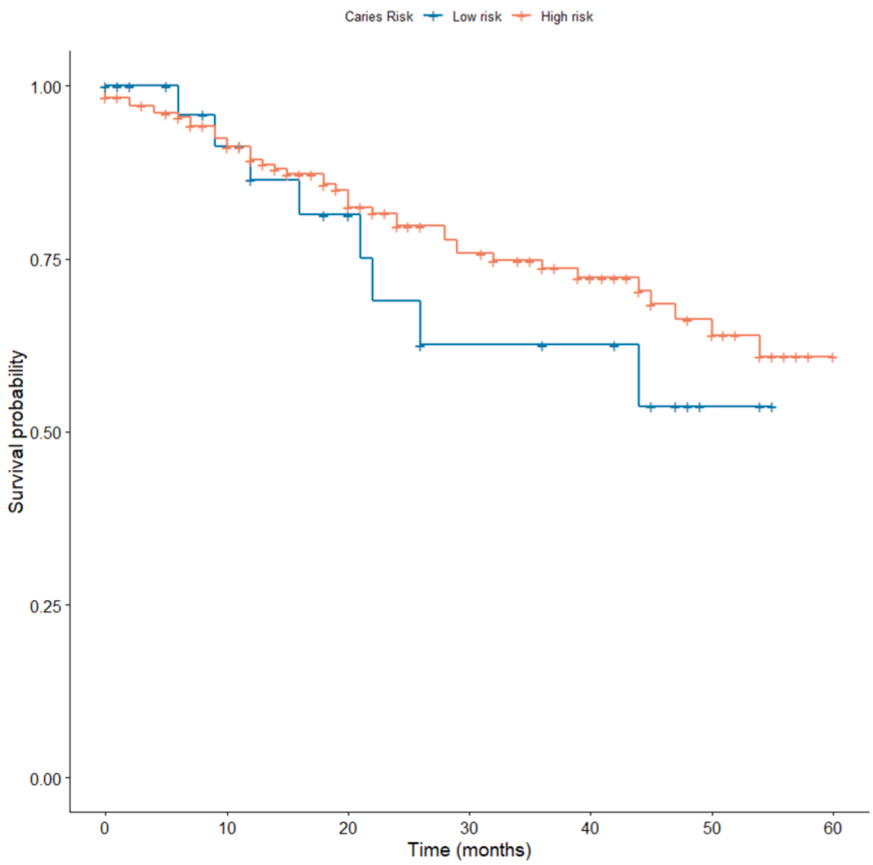

Fig. 4. Klapan-Meier survival estimates for proximal restorations grouped by restorative need of the children, during the 5 years of observation (p = 0.450).

patients, this age group also exhibited a significantly higher rate of restoration failure. So, we speculate that these younger patients, despite being sedated, they still can show uncooperative behaviour during treatment, probably due to their tender age and lack of understanding of their surroundings, which, in turn, lead to difficulties for the dentist to perform a proper isolation technique and reach the optimal conditions for the restorations. This may lead to an increased risk of salivary contamination, which may ultimately compromise restoration adhesion and survival. So, even occlusal restorations can show very low survival rates in our cohort. Additionally, such situations may call for GIC to be used, and we observed that, for these young patients with occlusal restoration, GIC also played an important role in restoration failure.

When adjusting the analyses in the multivariate model, some aspects stand out. We identified that cooperation and isolation were not significant any more. Actually, age was a more relevant factor. Again, the tender age of these patients explains their lack of cooperation, as well as the choice of isolation, so age is a far more relevant predictor that explains the model, and therefore remained significant in our analysis. Interestingly, occlusal restorations in older children had a markedly lower risk of failure. Patient maturity seems to play a strong role in restoration longevity. Another explanation for the risk of occlusal restoration failure may be related to caries risk. Once a cavitated lesion appears on the occlusal surface of a molar in such a young-aged patient, it may be speculated that the oral hygiene in these patients is inadequate and they have a higher risk of caries. So, secondary caries may develop on the margins of the restorations, or new caries lesions could develop in the proximal surfaces of the tooth, and the original restoration will have to be replaced. If, on the other hand, the first occlusal lesions develop only later in childhood (in older children), it is more probable that any further lesions will also take longer to develop, and thus, these occlusal restorations will have prolonged success rates.

Another significant factor associated with failure of occlusal restorations was position in the jaw. Restorations placed in the lower jaw had a significantly higher risk of failure, also likely due to the difficulty in performing good relative isolation with the movement of the tongue in these non-cooperative patients, thus increasing salivary contamination.

Proximal restorations showed considerably different characteristics. They were performed mostly in slightly older children, between 5 and 8 years old, under different sedation techniques (not only midazolam, but also nitrous oxide), and mostly performed under total isolation. In this age group, children may be expected to cooperate better under sedation, even though this variable was not significantly related to failure risk. Also, if the teeth survive the 5-year follow-up period examined in this study, they nearly reach the natural shedding age, and the restorations can then be considered successful. However, among the restorations that did fail, failure was rather related to the position of the tooth on the jaw (teeth on the lower jaw exhibited a significantly higher failure rate) and type of material. We speculate that the increased occlusal forces and masticatory stress, as well as greater chances of saliva accumulation on the lower jaw, may lead to lower survival of these restorations. Furthermore, proximal restorations made with GIC also had a threefold increase in the risk of failure, probably due to the lower mechanical strength and wear resistance when using this material on proximal surfaces. This is consistent with previous studies, where proximal GIC restorations have a fivefold greater risk of failure [22]. Our findings corroborate that, for proximal restorations, the interplay between the mechanical demands of the tooth location and the material will have a more crucial role than the patient-related factors.

When adjusting the results in the multivariate model, we clearly observe how the tooth-related variables are far more relevant than patient-related factors for proximal failures. Notably, patient-level variability played no role on the overall baseline model of survival. This reinforces the fact that older children tend to behave better under

Table 2
Bivariate and Multilevel analyses for proximal restorations.

Independent Variables	Restorations		Bivariate analyses		Multivariate analysis	
	Censored n (%)	Failure n (%)	Hazard ratio (95 %CI)	p-value	Hazard ratio (95 %CI)	p-value
Level 2						
Characteristics of the Child						
Sex						
Females	105 (36)	26 (9)				
Males	133 (46)	25 (9)	0.75(0.43–1.31)	0.316		
Age						
0–2	4 (1)	0 (0)				
3–4	38 (13)	12 (4)	638(0-~)	0.916		
5–6	112 (39)	27 (10)	510(0-~)	0.919		
7–8	74 (26)	12 (4)	285(0-~)	0.926		
9+	10 (3)	0 (0)	1.02(0-~)	1.000		
Restorative need						
Low	39 (83)	8 (17)				
High	199 (82)	43 (18)	0.76(0.32–1.77)	0.521		
Characteristics of the Treatment						
Sedation						
None	38 (17)	11 (5)				
N_2O/O_2	39(17)	11 (5)	1.24(0.54–2.86)	0.619		
Midazolam	101(44)	29 (13)	0.77(0.39–1.55)	0.470		
Cooperation						
Good	160 (55)	35 (12)				
Moderate	51 (18)	6 (2)	0.51(0.21–1.21)	0.124		
Bad	27 (10)	10 (3)	1.84(0.91–3.73)	0.091		
Level 1						
Characteristics of the Tooth						
Side of mouth						
Right	114 (39)	26 (9)				
Left	124 (43)	25 (9)	0.94(0.54–1.62)	0.816		
Jaw						
Upper	132 (46)	21 (7)				
Lower	106 (37)	30 (10)	1.91(1.09–3.31)	**0.024**	2.51(1.36–4.65)	**0.003**
Characteristics of the Restoration						
Isolation						
Cotton Roll	85 (29)	24 (8)				
Total isolation*	153 (53)	27 (9)	0.71(0.41–1.23)	0.217		
Restorative Material						
Composite	88 (30)	22 (8)				
Bulk fill	137 (47)	22 (8)	0.81(0.45–1.47)	0.491	0.89(0.47–1.68)	0.712
Glass Ionomer	13 (5)	7 (2)	3.60(1.52–8.49)	**0.003**	5.65(2.02–15.7)	**<0.001**

* Rubber dam or suction isolation technology.

Table 3
Reasons for failures in occlusal and proximal restorations.

Reason for failure	n (%) of occlusal	n (%) of proximal
Caries*	41 (77 %)	29 (57 %)
Pulpal involvement	2 (4 %)	9 (18 %)
Fracture / loss**	10 (19 %)	11 (22 %)
Other	–	2 (4 %)
Total	53 (100 %)	51 (100 %)

* Including secondary caries or new lesions forming on another surface of the same tooth that led to the need of a new restoration.
** 6 (out of the 10) occlusal restoration losses were from glass ionomer cements, whereas only 2 (out of the 11) proximal restoration losses were GIC.

sedation, so that patient-behaviour will play an irrelevant role on the survival. The explanatory power of the regression model was significantly improved when tooth-related variables were added. We observed that restorations in the lower jaw remained at a markedly higher risk of failure, with a twofold increase in hazard compared to upper jaw teeth. Similarly, GIC restorations exhibited a greater failure risk, with a fivefold increase in hazard. However, the number of GIC restorations (particularly Class II) was small, which may lead to unstable or inflated hazard ratios. Thus, the higher risk associated with GIC should be interpreted cautiously, as sparse data may overstate the true effect. The final model, which retained only jaw location and restorative material as predictors, seems to reiterate that occlusal forces and material properties are the main culprits in proximal restoration failures.

Analysing the reasons for restoration failures provides further insights into the mechanisms behind these findings. Similar to what has been reported in the literature [4,5], the risk of caries – either from new caries lesions or secondary caries – was the primary cause of failure, particularly in occlusal restorations. This is usually the case for high risk, young-aged children [23]. This emphasizes the theory that younger patients who needed occlusal restorations at early age will be more at risk of developing new lesions that ultimately require the replacement the original restorations. Also, the challenge in maintaining a dry field in these patients with poorer cooperation can lead to compromised adhesion and subsequent marginal breakdown of the restorations. In contrast, proximal restorations exhibited a higher incidence of pulpal involvement, which could indicate that the older children who have mostly proximal lesions could also have deeper cavities, which, in turn, are more at risk of bacterial contamination to the pulpal chamber. Interestingly, loss of restorations was observed in similar numbers in both occlusal and proximal restorations. However, the former was primarily linked to failures involving GIC, while the latter was mostly related to failures in the composite or bulk-fill materials.

Since we had a total of 151 patients in which a total of 517 teeth were treated, there were cases where both occlusal and proximal restorations were made on the same patient. In this case, we also performed an overall multilevel regression considering both restoration types. For this

Table 4
Multilevel analyses considering all patients and restorations of both types together.

Independent Variables	Multivariate model	
	Hazard ratio (95 %CI)	p-value
Level 2		
Characteristics of the Child		
Sex		
Females		
Males	0.77 (0.48 – 1.25)	0.290
Age		
0–2		
3–4	0.78 (0.29 – 2.09)	0.621
5–6	0.46 (0.16 – 1.34)	0.153
7–8	0.28 (0.09 – 0.95)	**0.041**
9+	0.00 (0.00 – inf)	0.999
Restorative need		
Low		
High	1.19(0.59–2.41)	0.629
Characteristics of the Treatment		
Sedation		
None		
N_2O/O_2	0.97 (0.39 – 2.46)	0.955
Midazolam	0.63 (0.34 – 1.17)	0.142
Cooperation		
Good		
Moderate	0.59 (0.31 – 1.09)	0.093
Bad	1.35 (0.76 – 2.41)	0.312
Level 1		
Characteristics of the Tooth		
Side of mouth		
Right		
Left	1.09 (0.72 – 1.64)	0.676
Jaw		
Upper		
Lower	2.08 (1.34 – 3.23)	**0.001**
Characteristics of the Restoration		
Isolation		
Cotton Roll		
Total isolation*	1.05 (0.61 – 1.79)	0.862
Restorative Material		
Composite		
Bulk fill	0.91 (0.55 – 1.51)	0.722
Glass Ionomer	3.84 (1.74 – 8.44)	**<0.001**
Location of Restoration		
Occlusal		
Proximal	1.34 (0.80 – 2.23)	0.269

overview, the overall risk of restoration failure seems to be multifactorial, with both patient-level and tooth-level covariates playing a role. Specifically, younger children tend to have higher failure rates. Again, this variable remained in the model, as it best explains the behavioural challenges of the patients and the difficulties in maintaining optimal isolation during treatment (cooperation had borderline significance). Restorations placed in the lower jaw are consistently at greater risk of failure, which corroborates other studies [24]. We can only speculate that this is likely due to variations in saliva exposure in this region. Finally, the choice of restorative material remains a very significant factor, with GIC demonstrating significantly higher failure rates across both restoration types.

Most remarkably, in this comprehensive model of both occlusal and proximal restorations, we observed that restoration type was not a significant factor in restoration failure (HR = 1.35, $p = 0.251$). In our cohort, all restorations had been placed in a university setting, by experienced and specialized dentists [15], which may explain the technical proficiency of the operators that allowed for the similar survival rates between occlusal and proximal restorations. We, therefore, speculate that, for patients who must be treated under sedation, as long as the treatment is performed by experienced pediatric dentists, the long-term success of the restorations will depend on specific patient- and tooth-related factors, namely: age, jaw position, and material used. Notably, the younger the patient is, with caries on their lower molars, and who are treated with GIC, the higher the risk of restoration failure. These findings underscore the need to prioritize these variables when planning restorative interventions.

There are several limitations to this study. An initial limitation is the setting of our clinic, since we are a referral clinic for difficult patients, only such patients are included in this study. Another important limitation is related to the operator bias. Studies have already shown that success rates of restorations are highly dependent on operators [25,26], where their experience seems to play a major role [8,27]. In our clinic, we have several treating dentists, which would lead to varied success rates, but our team is also composed of specialists, which may have led to a more standardized treatment, and in turn, improved success rates. This, however, remains only speculative, since we were not able to include the variable "operator experience" in our analyses. Another limitation is related to the baseline caries index of our patients. In this study we only focused on the cavitated lesions, and we, unfortunately, do not have any data on initial caries lesions on other surfaces, neither on the condition of the patient's oral hygiene. This could have provided a better understanding of the caries risk of our cohort. Still, the children who need sedation are those who present behavior problems and usually have dental fear [1], and, in turn, dental fear is associated with higher caries index [9]. So, it is plausible that our cohort had higher caries index in comparison with normal patients who are treated without sedation.

In this study, the clinical files did not systematically document the specific reasons for teeth lost-to-follow-up (e.g., exfoliation, end of follow-up, missed examinations, or drop-out patients). Consequently, all non-failure observations were treated as right-censored at the last recorded visit. Although this may introduce some risk of informative censoring, we found no evidence that censoring patterns differed between groups. While this is a limitation that should be acknowledged, the statistical models used (Kaplan–Meier and mixed-effects Cox regression) inherently account for right-censoring, and the distribution of censoring across study groups did not suggest systematic bias.

Overall, our analyses identified several predictors of failure, but their interpretation should be approached with caution. The higher failure rates observed with GIC restorations may be partly due to the small number of restorations placed with this material. Similarly, lower jaw restorations and younger patient age were associated with increased risk of failure, but these variables are also influenced by mechanical challenges, limited treatment time, and behavioral factors. So, it they do not have a straight-forward, causal effect. Furthermore, cooperation and isolation are clinically relevant variables, but they were not significant in our models. However, age seems to better explain these two variables (behavior and treatment conditions). While our study revealed predictors that may contribute to restoration failure, the clinical significance of our results should be interpreted in the context of a specialized pediatric clinic: one which primarily treats children with behavioral challenges and more complex clinical needs.

In conclusion, neither restoration type nor sedation method predicted restoration failures in our cohort, indicating that type of restoration (occlusal and proximal) or the type of sedation are not associated with restoration outcomes. Overall, restorations in non-compliant children are more likely to fail when placed in younger patients, in lower molars, or when glass ionomer cements are used.

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Thiago Saads Carvalho: Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization. **Chutima Un-Arm:** Writing – review & editing, Data curation. **Suvaniya**

Jeganathan: Writing – review & editing, Data curation. **Tiziana Hug:** Writing – review & editing, Methodology. **Richard Johannes Wierichs:** Writing – review & editing, Formal analysis. **Hendrik Meyer-Lueckel:** Writing – review & editing, Methodology. **Samira Helena Niemeyer:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors have nothing to declare.

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